



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

17-08-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	B	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	B,D	A,B,C,D	B,C	A,B,C	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.00	0.50	0.57	0.39	2.00	40.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	5	6			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	B	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C,D	A,C	B,C	A,B,C,D	B,D	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	2.00	20.00	80.00	18.00	-193.80
SECTION-II (ii)	Q.	7	8	9			
	A.	3	8	6			

PART-3 : MATHEMATICS

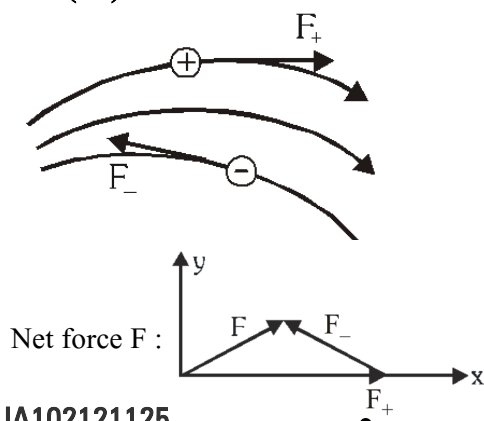
SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	C	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	A,B,D	A,B,D	A,C	C,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.40	1.57	30.41	21.42	2.25	15.75
SECTION-II (ii)	Q.	7	8	9			
	A.	2	10	7			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)



2. Ans (A)

$$R_{AB} = \int_0^{\ell} \frac{\rho_0}{A} \left(1 - \frac{x}{\ell}\right) dx = \frac{\rho_0}{2A} \ell$$

$$R_{BC} = \int_0^{\ell/3} \frac{\rho_0}{A} \left(1 - \frac{x}{\ell}\right) dx = \frac{5\rho_0 \ell}{18A}$$

$$\& R_{AC} = \frac{\rho_0 \ell}{2A} - \frac{5\rho_0 \ell}{18A} = \frac{2}{9} \frac{\rho_0 \ell}{A}$$

$$\text{at null point ; } \frac{X}{R} = \frac{R_{AC}}{R_{CB}} = \frac{2}{9} \times \frac{18}{5} = \frac{4}{5}$$

$$\therefore X = \frac{4R}{5}$$

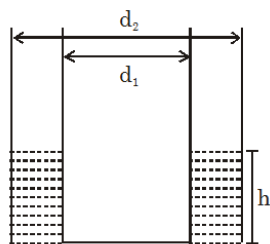
3. Ans (B)

$$\ell = 1.3 - 5 \times 0.01 = 1.25 \text{ cm}$$

$$y_C = \frac{m_1 \times 0 + m_2 \ell}{m_1 + m_2}$$

$$\begin{aligned} \frac{\Delta y_C}{y_C} &= \frac{\Delta m_2}{m_2} + \frac{\Delta \ell}{\ell} - \frac{(\Delta m_1 + \Delta m_2)}{m_1 + m_2} \\ &= \frac{\Delta m_2 m_1}{m_2 (m_1 + m_2)} + \frac{\Delta \ell}{\ell} + \frac{\Delta m_1}{m_1 + m_2} \\ &= \left(\frac{0.02 \times 1}{5 \times 4} + \frac{0.01}{1.25} + \frac{0.01}{5} \right) \times 100 \\ &= 0.1 + 0.8 + 0.2 = 1.1\% \end{aligned}$$

4. Ans (C)



Weight of water = Force due to surface tension.

$$mg = T\pi(d_2 + d_1)$$

$$\frac{\rho\pi(d_2^2 - d_1^2)hg}{4} = T\pi(d_2 + d_1)$$

$$\Rightarrow h = \frac{4T}{\rho g(d_2 - d_1)}$$

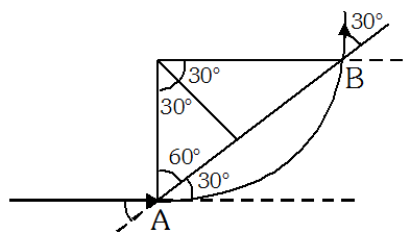
PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)

$$\text{Arc AB} = \frac{\pi}{3} r = \frac{\pi m V}{3qB}$$

$$\text{Time} = \left(\frac{1}{2\pi} \right) \left(\frac{\pi}{3} \right) = \frac{T}{6} = \frac{\pi m}{3qB}$$



6. Ans (B,D)

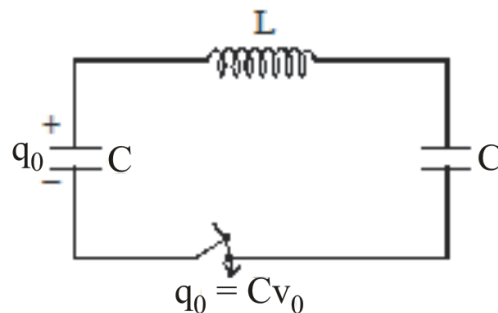
$$\rho gh = S \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$\rho gh = S \left(\frac{1}{r} + \frac{1}{\infty} \right)$$

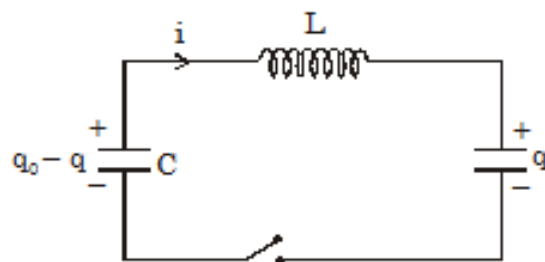
$$R = \frac{S}{\rho gh}$$

7. Ans (A,B,C,D)

At $t = 0$



At any time t



$$\text{Loop equation at } t \quad \frac{q_0 - q}{C} - L \frac{di}{dt} - \frac{q}{C} = 0$$

$$\Rightarrow L \frac{di}{dt} = \left(\frac{q_0}{C} - \frac{2q}{C} \right)$$

$$\Rightarrow \frac{d^2 q}{dt^2} = -\frac{2}{LC} \left(q - \frac{q_0}{2} \right)$$

Above equation is equation of SHM with

$$\omega = \sqrt{\frac{2}{LC}}$$

and solution of this equation will be

$$q - \frac{q_0}{2} = -\frac{q_0}{2} \cos \omega t \quad \{\text{As at } t = 0, q = 0\}$$

$$\Rightarrow q = \frac{q_0}{2} (1 - \cos \omega t) \quad i = \frac{q_0 \omega}{2} \sin \omega t$$

$$i_{\max} = \frac{q_0 \omega}{2} = v_0 \sqrt{\frac{C}{2L}}$$

$$\varepsilon = -\frac{d\phi}{dt} = -L \frac{di}{dt} = \frac{q_0 \omega^2 L}{2} \cos \omega t$$

$$\varepsilon_{\max} = \frac{q_0 \omega^2 L}{2} = v_0$$

Energy stored in inductor =

Initial energy in capacitor - Final energy in capacitors

$$= \frac{1}{2} \frac{q_0^2}{C} - \left(\frac{1}{2} \frac{(q_0/2)^2}{C} \times 2 \right) = \frac{1}{4} \frac{q_0^2}{C}$$

9. Ans (A,B,C)

$$I_{rms} = \frac{E_{rms}}{Z}$$

$$2.2 = \frac{220}{Z}$$

$$Z = 100 \Omega$$

$$\text{Hence } X_L = X_C$$

$$L\omega = \frac{1}{\omega C}$$

$$C = \frac{1}{\omega^2 L} = \frac{1}{(100\pi)^2 \cdot \frac{1}{\pi}} = \frac{1}{10^4 \pi} F = \frac{100}{\pi} \mu F$$

$$\therefore X_L = X_C$$

$$\text{Hence } \cos \phi = \frac{R}{C} = 1$$

$$P = E_{rms} I_{rms} = 220 \times 2.2 \times 10 = 484 \text{ W}$$

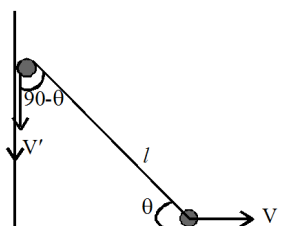
After inductor is short circuits impedance

$$Z = 100\sqrt{2}$$

$$\therefore I_{rms} = \frac{200}{100\sqrt{2}} = \left(\frac{2.2}{\sqrt{2}} \right)$$

10. Ans (A,B,C,D)

Obviously FBD of upper ball as tension becomes zero it loses contact, speed of lower ball maximize.



$$(1) V' \sin \theta = V \cos \theta$$

$$(2) \frac{1}{2} m V^2 + \frac{1}{2} m V'^2 = mg(\ell - \ell \sin \theta)$$

$$V^2 + V'^2 = 2gl(1 - \sin \theta)$$

$$V^2 + V'^2 \cot^2 \theta = 2gl(1 - \sin \theta)$$

$$V = \sqrt{\frac{2gl(1 - \sin \theta)}{\cos^2 \theta}} = \sin \theta \sqrt{2gl(1 - \sin \theta)}$$

$$\text{For max: } \frac{d}{d\theta} \sin^2 \theta (1 - \sin \theta) = 0$$

$$(1 - \sin \theta)(2 \sin \theta \cos \theta) + \sin^2 \theta (-\cos \theta) = 0$$

$$2 \cos \theta (1 - \sin \theta) = \sin \theta \cos \theta$$

$$2(1 - \sin \theta) = \sin \theta$$

$$\sin \theta = \frac{2}{3}$$

$$V = \frac{2}{3} \sqrt{2gl \cdot \frac{1}{3}}$$

$$= \sqrt{\frac{8gl}{27}} \text{ \& } V' = \sqrt{\frac{8gl}{27}} \times \sqrt{\frac{5}{4}} = \sqrt{\frac{10gl}{27}}$$

PART-1 : PHYSICS

SECTION-II (i)

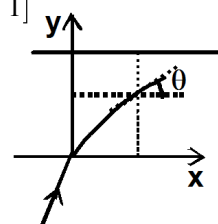
1. Ans (1.00)

$$\tan \theta = \frac{1}{\sqrt{\mu^2 - 1}} \quad [\because \mu \sin \theta = 1]$$

$$\therefore \frac{dy}{dx} = e^{-x/a}$$

$$\therefore y = a(1 - e^{-x/a})$$

$$\therefore \text{At } y = a/2, \quad x = a \ln 2$$



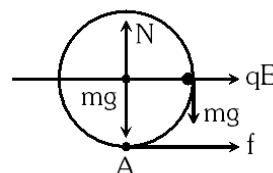
2. Ans (0.50)

$$\tan \theta = 1/2 \text{ at } x = a \ln 2$$

4. Ans (0.39)

$$\text{At } t = 0; qE + f = 2ma;$$

$$mgR + qER = I_A \alpha = I_A \frac{a}{R};$$



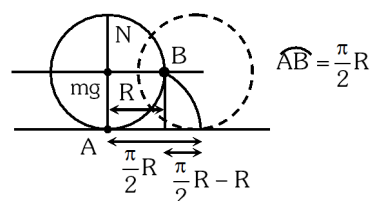
$$mg + qE = (mR^2 + mR^2 + 2mR^2) \frac{a}{R^2}$$

$$qE = mg, \text{ so } a = \frac{g}{2} \quad f = 2ma - qE = 2m \frac{g}{2} - mg = 0$$

Work done by electric field can be calculated by calculating.

Displacement of charge in the direction of field which is (see figure)

$$S = \left(\frac{\pi}{2} - 1 \right) R$$

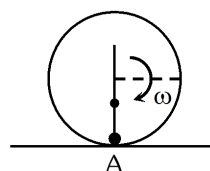


$$\text{So work done} = qER \left(\frac{\pi}{2} - 1 \right) = \frac{4}{7} mgR$$

After 90° rotation. Using work energy theorem

$$\frac{4}{7} mgR + mgR = \frac{1}{2} I_A \omega^2; \quad \sqrt{\frac{11}{7} \frac{g}{R}} = \omega$$

$$v_{com} = \frac{R}{2} \omega = \sqrt{\frac{11}{7} \frac{g}{R} \frac{R^2}{4}} = \sqrt{\frac{11}{28}} Rg$$



6. Ans (40.00)

$$Q = \frac{h^2}{24m\lambda^2} = K_B + K_C$$

where k is the kinetic energy

From conservation of linear momentum

$$|\vec{P}_C| = |\vec{P}_B| = P$$

$$\therefore K = \frac{P^2}{2m} \therefore Q = \frac{P^2}{2 \times 12m} + \frac{P^2}{2 \times 4m}$$

$$\frac{h^2}{24m\lambda^2} = \frac{P^2}{2} \left(\frac{1}{12m} + \frac{1}{4m} \right)$$

$$\frac{h^2}{12m\lambda^2} = P^2 \left(\frac{1+3}{12m} \right) = \frac{P^2}{3m}$$

$$\Rightarrow P^2 = \frac{h^2 \times 3m}{12m\lambda^2} \Rightarrow P = \frac{h}{2\lambda}$$

$$\Rightarrow \frac{h}{P} = 2\lambda = \text{de Broglie wavelength}$$

$$\therefore |\vec{P}_B| = |\vec{P}_C|$$

$$\text{Mass defect, } \Delta m = \frac{Q}{C^2} = \frac{h^2}{24mc^2\lambda^2}$$

$$\therefore m_A = 16m + \frac{h^2}{24mc^2\lambda^2}$$

PART-1 : PHYSICS

SECTION-II (ii)

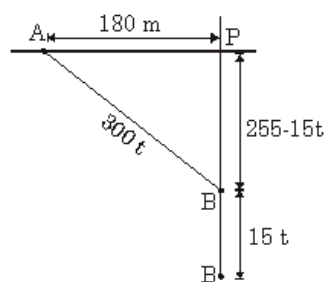
7. Ans (3)

$$(180)^2 + (255 - 15t)^2 = (300t)^2$$

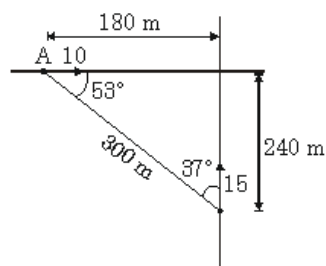
$$32400 + 65025 + 225t^2 - 7650t = 90000t^2$$

$$89775t^2 + 7650t - 97425 = 0$$

$$\Rightarrow t = 1 \text{ sec}$$



$$f = \left(\frac{300 + 15 \cos 37}{300 - 10 \cos 53} \right) f_0 = \left(\frac{300 + 12}{300 - 6} \right) f_0$$



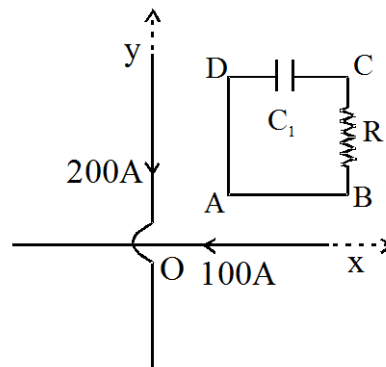
$$= \frac{312}{294} \times 2.94 \text{ kHz}$$

$$f = 3.12 \text{ kHz}$$

8. Ans (5)

The emf induced in the anticlockwise direction in the loop is

$$\frac{\mu_0 100}{2\pi \times 2 \times 0.1} \times 0.1 \times 30 - \frac{\mu_0 100}{2\pi \times 3 \times 0.1} \times 0.1 \times 30$$



$$= \frac{\mu_0}{2\pi} \times 100 \times 30 \left(\frac{1}{2} - \frac{1}{3} \right) = 10^{-4} \text{ V}$$

$\therefore$ The net emf is anticlockwise and is given by 10^{-4} V .

Apply kirchoff law along the loop.

$$+\frac{q}{C} + iR = 10^{-4}$$

$$i = \left[10^{-4} - \frac{q}{C} \right] \frac{1}{R} = 10^{-5} \text{ A.}$$

9. Ans (6)

$$Q = mL_f$$

$$\text{Thickness } L = \frac{tkA [T_H - T_C]}{Q} = 6$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)

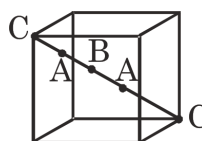
$$\sqrt{3}a = r_c + 2r_A + 2r_B + 2r_A + r_c$$

$$= 2r_C + 4r_A + r_B = 2r_C + 4 \times 0.225 r_C + 2 \times 0.414 r_C$$

$$= 3.728 r_C$$

$$\sqrt{2}a = 4r_C$$

$$\text{Fraction occupied} = \frac{3.728 r_C}{\sqrt{3} \times \frac{4r_C}{\sqrt{2}}} = 0.7609$$



Fraction of body diagonal not covered = 0.24

2. Ans (C)

Probability of finding the electron is

$$\psi^2 = 0 \quad \text{or} \quad \psi = 0 \quad \sigma = \frac{2Zr}{a_0}$$

$$(\sigma - 1) = 0 \Rightarrow \sigma = 1 \quad r = \frac{\sigma a_0}{2Z}$$

$$\text{or } r = \frac{a_0}{2Z} \quad \text{or } \sigma^2 - 8\sigma + 12 = 0$$

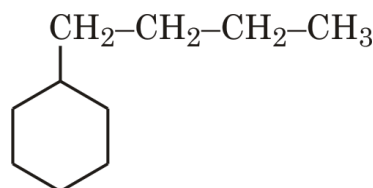
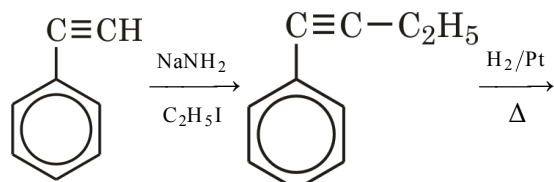
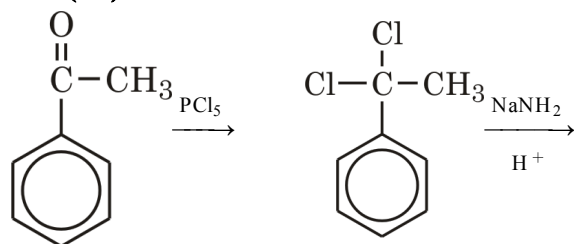
$$\sigma = 6; \quad \sigma = 2$$

$$\text{if } \sigma = 6 \Rightarrow r = \frac{6a_0}{2Z} = \boxed{\frac{3a_0}{Z}}$$

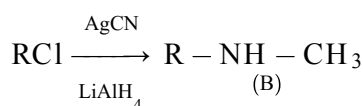
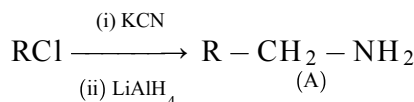
$$\sigma = 2 \quad r = \frac{2a_0}{2Z} = \frac{a_0}{Z}$$

$$\sigma = 1 \quad r = \frac{a_0}{2Z} = \boxed{\frac{a_0}{2Z}}$$

3. Ans (B)



4. Ans (C)

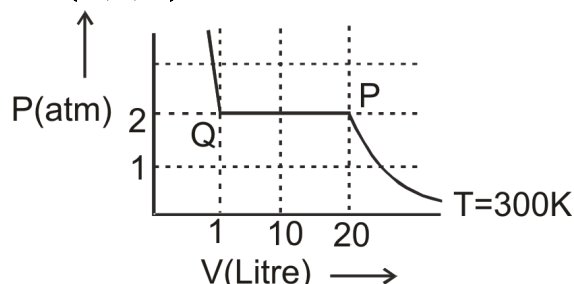


A & B are functional isomers.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,C,D)



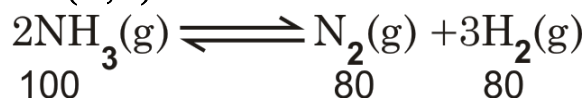
(A) A real gas condenses if external pressure is just greater than vapour pressure (B,C,D)

At point 'P' whole real gas is in gaseous phase
so density of gas = $\frac{1000}{20 \times 1000} = 0.05 \text{ gm/ml}$

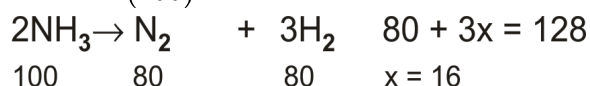
At point 'Q' whole substance is in liquid phase
so density of liquid = $\frac{1000}{1 \times 1000} = 1 \text{ gm/ml}$

Note : during condensation at 300K density of gas and liquid remains constant, only amount varies.

6. Ans (A,C)



$$K_P = \frac{80 \times (80)^3}{(100)^2}$$



$$100 \quad 80 \quad 80 \quad x = 16$$

$$100 \quad 80-p \quad 80-p$$

$$100-2x \quad 80-p+x \quad 80+3x$$

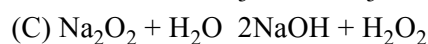
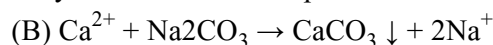
$$100 \quad 96-p \quad 128$$

$$K_P = \frac{(96-p) \times (128)^3}{(68)^2} = \frac{(80)^4}{(100)^2}$$

$$(96-p) = 9 = \text{Partial pressure of } \text{N}_2.$$

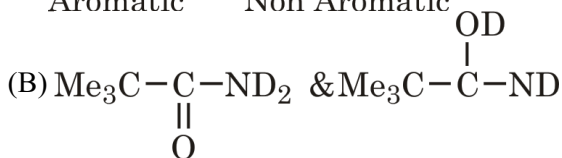
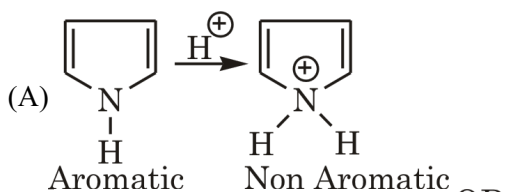
7. Ans (B,C)

(A) Around $T = 300 \text{ K}$, gypsum is not dehydrated to Plaster of Paris

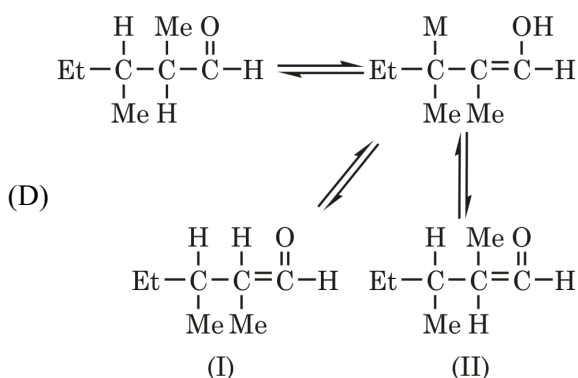


(D) $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$ mobility in aqueous solution.

8. Ans (A,B,C,D)



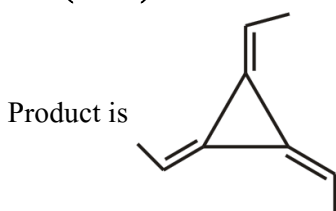
(C) CH_3S^- is more stable than CH_3O^- due to larger size of sulphur atom



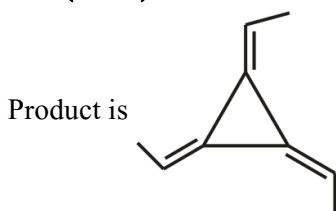
PART-2 : CHEMISTRY

SECTION-II (i)

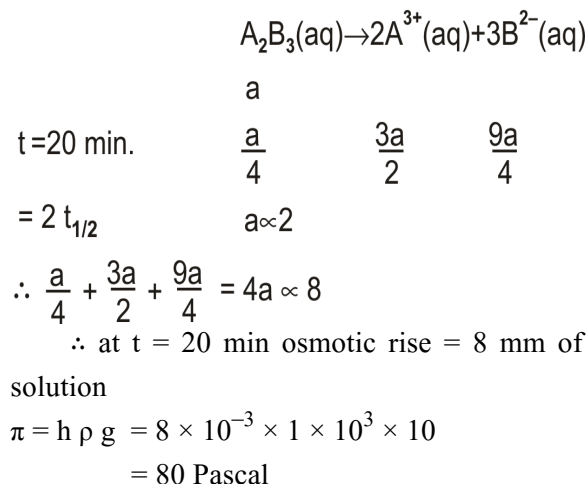
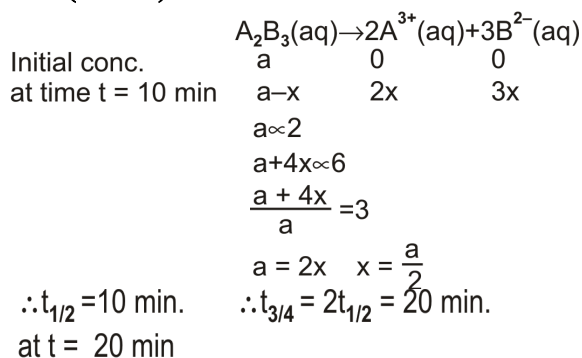
1. Ans (2.00)



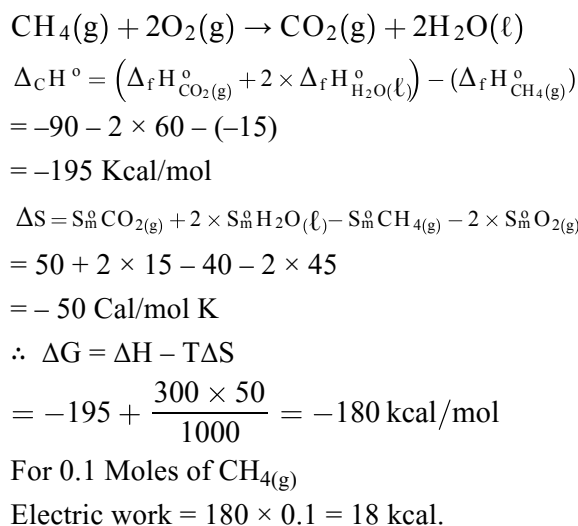
2. Ans (2.00)



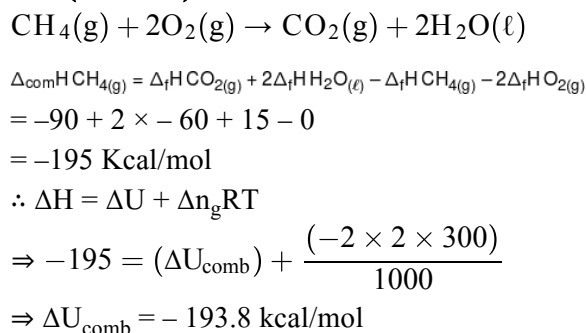
4. Ans (80.00)



5. Ans (18.00)



6. Ans (-193.80)



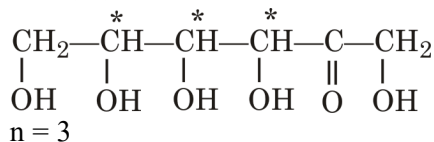
PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (3)**

(i), (iii) and (v) are not aromatic.

8. **Ans (8)**



Total stereoisomers = $2^3 = 8$

9. **Ans (6)**

$$\Delta T_f = i \times K_f m$$

$$K_{sp} = 4s^3 = 4 \times 10^{-6}$$

$$s = 10^{-2} \text{M}$$

$$\Delta T_f = 3 \times 2 \times 10^{-2}$$

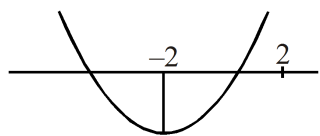
$$= \frac{6}{100}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

$$x^4 + kx^3 + x^2 + kx + 1 = 0$$



$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right) + k \left(x + \frac{1}{x}\right) + 1 = 0$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 + k \left(x + \frac{1}{x}\right) - 1 = 0$$

$$\text{Let } x + \frac{1}{x} = t$$

$$t^2 + kt - 1 = 0$$

has exactly one root less than -2 & other root is

less than 2

$$f(t) = t^2 + kt - 1$$

$$\Rightarrow f(-2) < 0 \text{ \& \; } f(2) > 0$$

$$\Rightarrow 4 - 2k - 1 < 0$$

$$k > \frac{3}{2}$$

2. **Ans (B)**

$$\lim_{x \rightarrow 0} \frac{(1 + 5x + 4x^2)^{1/x} - (1 + 5x - 4x^2)^{1/x}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x} \ln(1+5x+4x^2)} - e^5}{x} - \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x} \ln(1+5x-4x^2)} - e^5}{x}$$

$$= L_1 - L_2$$

$$L_1 = \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x} \left[(5x+4x^2) - \frac{x^2(5+4x)^2}{2} + \dots \right]} - e^5}{x}$$

$$L_1 = \lim_{x \rightarrow 0} \frac{e^{5 + \left(4 - \frac{25}{2}\right)x + \dots} - e^5}{x}$$

$$= \lim_{x \rightarrow 0} \frac{e^5 \left[e^{-\frac{17}{2}x + \dots} - 1 \right]}{\frac{-17x}{2}} \times \left(-\frac{17}{2} \right)$$

$$L_1 = e^5 \left(-\frac{17}{2} \right)$$

Similarly

$$L_2 = e^5 \left(-4 - \frac{25}{2} \right) = -\frac{33}{2} e^5$$

$$L_1 - L_2 = e^5 \left[\frac{-17 + 33}{2} \right] = 8e^5$$

3. **Ans (C)**

(i) 4 4 1 1 1 1 1

$$\text{No. of ways} = \frac{7!}{2! \times 5!} = 21$$

(ii) 4 3 2 1 1 1 1

$$\text{No. of ways} = \frac{7!}{4!} = 7 \times 6 \times 5 = 210$$

(iii) 4 2 2 2 1 1 1

$$\text{No. of ways} = \frac{7!}{3! \cdot 3!} = \frac{7 \times 6 \times 5 \times 4}{3!} = 7 \times 5 \times 4 = 140$$

$$\text{Total number of ways} = 21 + 210 + 140 = 371$$

4. **Ans (D)**

$$g(x) = f(\tan^2 x + 2 \tan x + 2), x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

$$g'(x) = f'(\tan^2 x + 2 \tan x + 2) (2 \tan x + 2) \sec^2 x > 0$$

$$\Rightarrow f'((1 + \tan x)^2 + 1) 2(1 + \tan x) \sec^2 x > 0$$

$$\Rightarrow 1 + \tan x > 0$$

$$\tan x > -1$$

$$\Rightarrow -\frac{\pi}{4} < x < \frac{\pi}{2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,D)

Let circle S be $x^2 + y^2 + 2gx + 2fy + c = 0$

$(-g, -f)$ lies on $2x - 2y + 7 = 0$

$$\Rightarrow -2g + 2f + 7 = 0$$

$$2g = 7 + 2f$$

$S = 0$ cuts $x^2 + y^2 = 5$ orthogonally

$$\Rightarrow c = 5$$

$$\Rightarrow x^2 + y^2 + (2f + 7)x + 2fy + 5 = 0$$

$$x^2 + y^2 + 7x + 5 + 2f(x + y) = 0$$

which is a family of circle passing through

points of intersection of circle

$$x^2 + y^2 + 7x + 5 = 0 \text{ \& } x + y = 0$$

$$\Rightarrow 2x^2 + 7x + 5 = 0 \Rightarrow x = -1, -\frac{5}{2}$$

$$\Rightarrow \text{fixed points are } (-1, 1) \text{ \& } \left(-\frac{5}{2}, \frac{5}{2}\right)$$

6. Ans (A,B,D)

$$f'(x) = \begin{cases} -\sin x & x < 0 \\ 2x - 1 & x > 0 \end{cases}$$

$$f'(0^-) = 0, f'(0^+) = -1$$

$\Rightarrow f(x)$ is non differentiable at $x = 0$

$f(x)$ is differentiable at $x = 1$ and 2

$$\Rightarrow \frac{d}{dx}(f(f(x))) = f'(f(x)) \cdot f'(x)$$

at $x = 1$

$$= f'(f(1)) \cdot f'(1)$$

$$= f'(2) \cdot f'(1)$$

$$= 3$$

7. Ans (A,B,D)

$$|P| = 3, P^2 (\text{adj } P) = \begin{bmatrix} 3 & 3 & 0 \\ 0 & 3 & 3 \\ k & 0 & 3 \end{bmatrix}$$

$$\Rightarrow P(P \text{ adj } P) = \begin{bmatrix} 3 & 3 & 0 \\ 0 & 3 & 3 \\ k & 0 & 3 \end{bmatrix}$$

$$P(|P| I) = \begin{bmatrix} 3 & 3 & 0 \\ 0 & 3 & 3 \\ k & 0 & 3 \end{bmatrix}$$

$$||P| P| = \begin{bmatrix} 3 & 3 & 0 \\ 0 & 3 & 3 \\ k & 0 & 3 \end{bmatrix}$$

$$|P|^4 = 4(9 + 3k)$$

$$\Rightarrow 3^4 = 3(9 + 3k)$$

$$\Rightarrow k = 6$$

$$\Rightarrow 3P = \begin{bmatrix} 3 & 3 & 0 \\ 0 & 3 & 3 \\ 6 & 0 & 3 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\Rightarrow \text{tr}(P^2) = 3$$

$$\text{tr}(\text{adj } P) = 3$$

$$|\text{adj } 2P^{-1}| = |2P^{-1}|^2$$

$$= 2^6 |P^{-1}|^2 = \frac{2^6}{|P|^2} = \frac{64}{9}$$

$$|\text{adj } (2\text{adj } P)| = |2\text{adj } P|^2$$

$$= 2^6 |\text{adj } P|^2$$

$$= 2^6 |P|^4$$

$$= 64 \times 81 = 5184$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (0.40)**

$$f(x) + f(1-x) = 1$$

$$I = \int_{1/10}^{9/10} f(f(x)) dx = \int_{1/10}^{9/10} f(f(1-x)) dx$$

$$= \int_{1/10}^{9/10} f(1-f(x)) dx$$

$$I = \int_{1/10}^{9/10} (1-f(f(x))) dx$$

$$\Rightarrow 2I = \int_{1/10}^{9/10} 1 dx \Rightarrow 2I = \frac{8}{10}$$

$$I = 0.40$$

2. **Ans (1.57)**

$$\text{Let } g(\theta) = S$$

$$S = 1 + (1 + \sin\theta)\cos\theta + (1 + \sin\theta + \sin^2\theta)\cos^2\theta + \dots \infty$$

$$S \cos\theta = \cos\theta + (1 + \sin\theta)\cos^2\theta + \dots \infty$$

$$S(1 - \cos\theta) = 1 + \sin\theta \cos\theta + \sin^2\theta \cos^2\theta + \dots \infty$$

$$S = \frac{1}{(1 - \sin\theta \cos\theta)(1 - \cos\theta)}$$

$$I = \int_0^{\pi/2} \frac{(1 - \cos\theta) \cos\theta}{(1 - \sin\theta \cos\theta)(1 - \cos\theta)} d\theta$$

$$I = \int_0^{\pi/2} \frac{\cos\theta}{1 - \sin\theta \cos\theta} d\theta$$

$$I = \int_0^{\pi/2} \frac{2 \cos\theta}{2 - \sin 2\theta} d\theta \quad \dots(1)$$

$$I = \int_0^{\pi/2} \frac{2 \sin\theta}{2 - \sin 2\theta} d\theta \quad \dots(2)$$

adding (1) & (2)

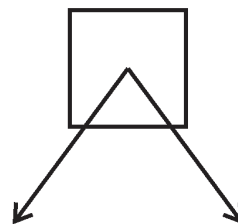
$$2I = \int_0^{\pi/2} \frac{2(\sin\theta + \cos\theta)}{2 - \sin 2\theta} d\theta$$

$$I = \int_0^{\pi/2} \frac{(\cos\theta + \sin\theta) d\theta}{2 - (1 - (\sin\theta - \cos\theta)^2)} \quad \text{let } \sin\theta - \cos\theta = t$$

$$I = \int_{-1}^1 \frac{dt}{1+t^2} \Rightarrow I = 2 \int_0^1 \frac{dt}{1+t^2}$$

$$I = 2 [\tan^{-1}t]_0^1 \Rightarrow I = 2 \left[\frac{\pi}{4} \right] = \frac{\pi}{2} = \frac{3.14}{2} = 1.57$$

3. **Ans (30.41)**



3 distinct dice 5 distinct dice

$$1, 2, 3 \rightarrow 3! = 6 \quad 1, 1, 1, 1, 2 = \frac{5!}{4!} = 5$$

$$1, 1, 4 \rightarrow \frac{3!}{2!} = 3$$

$$2, 2, 2 \rightarrow 1$$

$$P_1 = \frac{1}{2} \times \frac{10}{6^3} + \frac{1}{2} \times \frac{5}{6^5}$$

$$P_1 = \frac{5}{6^3} + \frac{5}{2 \times 6^5}$$

$$1296P_1 = 5 \times 6 + \frac{5}{2 \times 6}$$

$$= 30 + \frac{5}{12} = 30.41$$

4. **Ans (21.42)**

3 dice 5 dice

$${}^6C_3 \times 3! \quad {}^6C_5 \times 5!$$

$$= 5! \quad = 6 \times 5!$$

$$P_2 = \frac{\frac{1}{2} \times \frac{6 \times 5!}{6^5}}{\frac{1}{2} \times \frac{5!}{6^3} + \frac{1}{2} \times \frac{6 \times 5!}{6^5}}$$

$$= \frac{1}{7} \Rightarrow 150P_2 = 21.42$$

5. Ans (2.25)

$$\frac{x^2}{16} + \frac{y^2}{12} = 1$$

$$e = \sqrt{1 - \frac{12}{16}} = \frac{1}{2}$$

foci of ellipse are $(\pm 2, 0)$

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is passing through $(\pm 2, 0)$

$$\Rightarrow \frac{4}{a^2} = 1 \Rightarrow a^2 = 4$$

eccentricity of hyperbola = 2

$$\Rightarrow \sqrt{1 + \frac{b^2}{a^2}} = 2$$

$$\Rightarrow b^2 = 12$$

$$\Rightarrow \text{hyperbola will be } \frac{x^2}{4} - \frac{y^2}{12} = 1$$

end point of L.R. $\left(ae, \frac{b^2}{a} \right)$

$$\left(2, \frac{12}{4} \right) \Rightarrow (2, 3)$$

tangent to ellipse at $(2, 3)$

$$\frac{x}{8} + \frac{y}{4} = 1 \Rightarrow x + 2y - 8 = 0$$

equation of required circle will be

$$(x-2)^2 + (y-3)^2 + \lambda(x+2y-8) = 0$$

it passing through $(0, 0)$

$$\Rightarrow \lambda = \frac{13}{8}$$

$$\Rightarrow (x-2)^2 + (y-3)^2 + \frac{13}{8}(x+2y-8) = 0$$

$$\Rightarrow 8[x^2 + y^2 - 4x - 6y + 13] + 13x + 26y - 13 \times 8 = 0$$

$$\Rightarrow 8x^2 + 8y^2 - 19x - 22y = 0$$

$$x^2 + y^2 - \frac{19}{8}x - \frac{22}{8}y = 0$$

$$\Rightarrow g = -\frac{19}{16}, f = -\frac{22}{16}$$

$$\Rightarrow 12(g-f) = \frac{9}{4} = 2.25$$

6. Ans (15.75)

$$\frac{x^2}{16} + \frac{y^2}{12} = 1$$

$$e = \sqrt{1 - \frac{12}{16}} = \frac{1}{2}$$

foci of ellipse are $(\pm 2, 0)$

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is passing through $(\pm 2, 0)$

$$\Rightarrow \frac{4}{a^2} = 1 \Rightarrow a^2 = 4$$

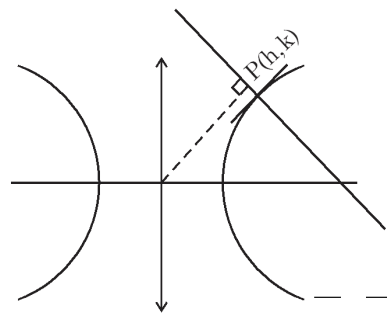
eccentricity of hyperbola = 2

$$\Rightarrow \sqrt{1 + \frac{b^2}{a^2}} = 2$$

$$\Rightarrow b^2 = 12$$

$$\Rightarrow \text{hyperbola will be } \frac{x^2}{4} - \frac{y^2}{12} = 1$$

Let equation of normal be $y = mx \pm \frac{m(a^2 + b^2)}{\sqrt{a^2 - m^2b^2}}$



$$\Rightarrow y = mx \pm \frac{16m}{\sqrt{4 - 12m^2}}$$

$$\text{Let } P(h, k) \Rightarrow k = mh \pm \frac{8m}{\sqrt{1 - 3m^2}}$$

$$(m) \left(\frac{k}{h} \right) = -1 \Rightarrow m = -\frac{h}{k}$$

$$\Rightarrow k = -\frac{h^2}{k} \pm \frac{\left(\frac{-8h}{k} \right)}{\sqrt{1 - \frac{3^2}{k^2}}}$$

$$\frac{k^2 + h^2}{k} = \mp \frac{8h}{\sqrt{k^2 - 3h^2}}$$

$$\Rightarrow \frac{(k^2 + h^2)^2}{k^2} = \frac{64h^2}{(k^2 - 3h^2)}$$

$$\Rightarrow (x^2 + y^2)^2 (y^2 - 3x^2) = 64x^2y^2$$

$$\Rightarrow (3x^2 - y^2)(x^2 + y^2)^2 + 64x^2y^2 = 0$$

$$\Rightarrow (3x^2 + Ay^2)(x^2 + y^2)^2 + Bx^2y^2 = 0$$

$$\Rightarrow A = -1, B = 64$$

$$\frac{A+B}{4} = \frac{63}{4} = 15.75$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (2)

$$2 \int_1^x f(t) dt + x^{k+1} = xf(x)$$

$$2f(x) + (k+1)x^k = f(x) + xf'(x)$$

$$\Rightarrow xf'(x) - f(x) = (k+1)x^k$$

$$\Rightarrow f'(x) - \frac{f(x)}{x} = (k+1)x^{k-1}$$

$$\text{I.F.} = e^{\int -\frac{1}{x} dx} = \frac{1}{x}$$

$$f(x) \cdot \frac{1}{x} = \int (k+1)x^{k-2} dx$$

$$\frac{f(x)}{x} = (k+1) \frac{x^{k-1}}{k-1} + c$$

$$\text{put } x = 1$$

$$f(1) = \frac{k+1}{k-1} + c$$

$$1 = \frac{k+1}{k-1} + c$$

$$\Rightarrow c = 1 - \frac{k+1}{k-1}$$

$$c = \frac{-2}{k-1} = \frac{2}{1-k}$$

$$\Rightarrow f(x) = \frac{k+1}{k-1} x^k + \frac{2}{1-k} x$$

$$\frac{k+1}{k-1} \& \frac{2}{1-k} \text{ must be integer}$$

$$\Rightarrow \frac{k+1}{k-1} = 1 + \frac{2}{1-k}$$

$$\Rightarrow \frac{2}{1-k} \text{ must be integer}$$

$$1-k = 1, -1, 2, -2$$

$$-k = 0, -2, 1, -3$$

$$k = 0, 2, -1, 3$$

$$\Rightarrow \text{possible values of } k = 2, 3$$

8. Ans (10)

$$\frac{1}{a+5} + \frac{1}{b+5} = \frac{1}{5} - \frac{1}{c+5}$$

$$\frac{1}{a+5} + \frac{1}{b+5} = \frac{c}{5(c+5)}$$

$$\text{apply A.M.} \geq \text{G.M. in } \frac{1}{a+5} \& \frac{1}{b+5}$$

$$\frac{1}{a+5} + \frac{1}{b+5} \geq 2 \left(\frac{1}{(a+5)} \cdot \frac{1}{(b+5)} \right)^{1/2}$$

$$\Rightarrow \frac{c}{5(c+5)} \geq 2 \left(\frac{1}{(a+5)} \cdot \frac{1}{(b+5)} \right)^{1/2}$$

similarly

$$\frac{b}{5(b+5)} \geq 2 \left(\frac{1}{(a+5)} \cdot \frac{1}{(c+5)} \right)^{1/2}$$

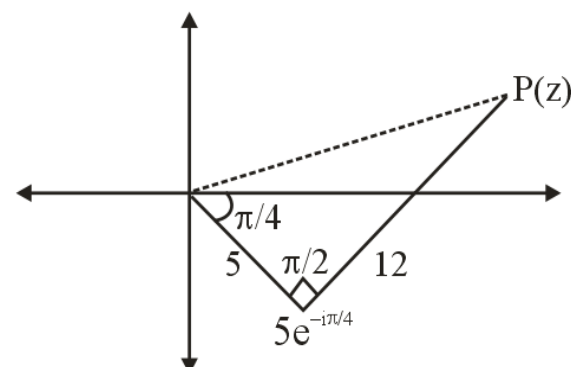
$$\& \frac{a}{5(a+5)} \geq 2 \left(\frac{1}{(b+5)} \cdot \frac{1}{(c+5)} \right)^{1/2}$$

on multiplying

$$\frac{abc}{5^3(a+5)(b+5)(c+5)} \geq \frac{8}{(a+5)(b+5)(c+5)}$$

$$\Rightarrow \frac{abc}{5^3} \geq 8 \Rightarrow (abc)^{1/3} \geq 10$$

9. Ans (7)



$$\frac{z - 5e^{-i\pi/4}}{12} = \left(\frac{-5e^{-i\pi/4}}{5} \right) e^{-i\pi/2}$$

$$z - 5e^{-i\pi/4} = 12ie^{-i\pi/4}$$

$$z = (5 + 12i) e^{-i\pi/4} = (12 - 5i) e^{i\pi/4}$$

$$\Rightarrow a = 12, b = -5$$

$$[a + b] = 7$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

17-08-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B	C,D	A,B,C,D	B,C	A,C,D	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	B	B	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.80	0.92	56.00	56.00	33.33	2250.00
SECTION-II (ii)	Q.	7	8	9			
	A.	20	6045	4			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C,D	A,C	A,B,C,D	B,C,D	B,C,D	C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	D	B	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	564.70 to 565.00	675.00	4.00	4.00	13.00	1.00
SECTION-II (ii)	Q.	7	8	9			
	A.	335	6	8			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,D	A,B,C	B,D	B,C	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	C	C	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	47.00	34.00	1.50	1.00	0.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	683	5			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B)

$$V_1 = 50$$

$$V_2 = 10t$$

Initially 1 will be ahead & finally two will overtake 1

$$\text{So } 125 = 50t_1 - \frac{1}{2} \times 10 \times t_1^2 \dots (i)$$

$$\& 125 = \frac{1}{2} \times 10t_2^2 - 50t_2 \dots (ii)$$

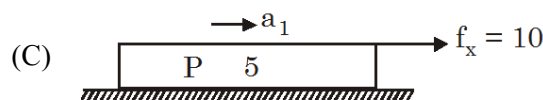
on solving (i)

$$t_1 = 5$$

on solving (ii)

$$t_2 = 5(1 + \sqrt{2})$$

2. Ans (C,D)



$$10 = 5a,$$

$$\therefore a_1 = 2$$

$$(D) 15 = (2 + 5)a_c$$

3. **Ans (A,B,C,D)**

(i) Velocity of flow

$$= \sqrt{2gh} = \sqrt{2 \times 10 \times 3.6} = 6\sqrt{2} \text{ m/s}$$

(ii) Rate of flow $= Av = \pi \left(\frac{4 \times 10^{-2}}{\sqrt{\pi}} \right)^2 \times 6\sqrt{2}$

$$= 9.6 \times \sqrt{2} \times 10^{-3} \text{ m}^3/\text{s}$$

(iii) Bernoulli's theorem between surface and A

$$P_{\text{atm}} = P + \frac{1}{2} \rho v^2 + \rho gh$$

$$\Rightarrow P = P_{\text{atm}} - \frac{1}{2} \rho v^2 - \rho gh$$

$$= 10^5 - \frac{1}{2} \times 10^3 (6\sqrt{2})^2 - 10^3 \times 10 \times 1.8$$

$$= 4.6 \times 10^4 \text{ N/m}^2$$

4. **Ans (B,C)**

Work done by the spring on block = loss in spring P.E.

$$= \frac{1}{2} ka^2 - \frac{1}{2} kb^2 = \frac{1}{2} k(a^2 - b^2)$$

this is also work done against friction $= \mu mg(a + b)$

$$\therefore \mu = \frac{\frac{1}{2} k(a^2 - b^2)}{mg(a + b)} = \frac{k}{2mg} (a - b)$$

6. **Ans (A,C)**

$$C = \frac{2\pi ab\epsilon_0 (k + 1)}{(b - a)}$$

$$R = \frac{(b - a)}{2\pi\sigma ab}$$

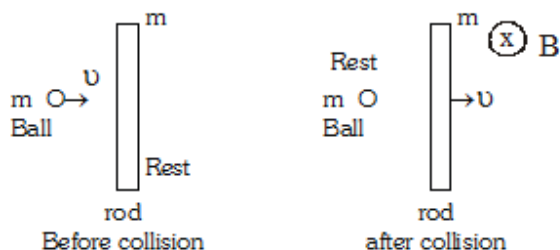
$$RC = \frac{2\pi ab\epsilon_0 (k + 1)}{(b - a)} \cdot \frac{(b - a)}{2\pi\sigma ab}$$

PART-1 : PHYSICS

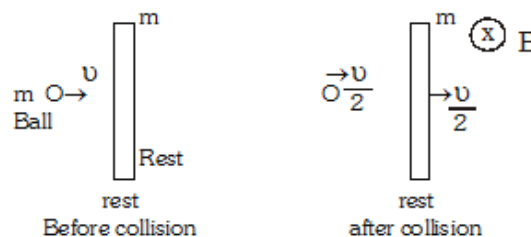
SECTION-I (ii)

8. **Ans (B)**

with $e = 1$



with $e = 0$



$$\Rightarrow \text{Motional EMF} = B\ell v$$

$$\Rightarrow \text{Motional EMF} = B\ell v/2$$

10. **Ans (C)**

For process AB

$$Q = nC_V \Delta T + W$$

$$= \frac{f}{2} \cdot (nR\Delta T) + \frac{1}{2} (P_0 + 4P_0) (3V_0)$$

$$Q = \frac{f}{2} \cdot (16P_0 V_0 - P_0 V_0) + \frac{15}{2} P_0 V_0$$

$$Q = (f + 1) \frac{15}{2} P_0 V_0 = 360 \dots (1)$$

Now for process C $\rightarrow$ A :

$$Q = \left(\frac{f}{2} + 1 \right) \cdot (3P_0 V_0 - P_0 V_0)$$

$$= \left(\frac{f}{2} + 1 \right) (2P_0 V_0) \dots (2)$$

Now

$$\frac{(1)}{(2)} = \frac{360}{56} = \frac{(f + 1) \frac{15}{2} P_0 V_0}{\left(\frac{f + 2}{2} \right) 2P_0 V_0} = \left(\frac{f + 1}{f + 2} \right) \cdot \frac{15}{2}$$

$$\Rightarrow \frac{45 \times 2}{7 \times 15} = \frac{f + 1}{f + 2} = \frac{90}{105} \cdot \frac{6}{7}$$

$$\Rightarrow f = 5 \quad \text{and}$$

$$(5 + 1) \frac{15}{2} P_0 V_0 = 360 \Rightarrow P_0 V_0 = 8$$

In process B $\rightarrow$ C

Heat rejected = Q

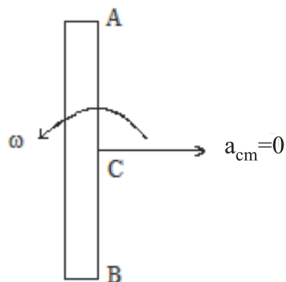
$$= \frac{f}{2} \cdot (16P_0 V_0 - 3P_0 V_0) + \frac{1}{2} \cdot (P_0 + 4P_0) (V_0)$$

$$= \frac{5}{2} \cdot 13P_0 V_0 + \frac{5}{2} \cdot P_0 V_0 = 35P_0 V_0 = 280 \text{ J}$$

PART-1 : PHYSICS

SECTION-II (i)

4. Ans (56.00)



$$\omega = \frac{11 + 3}{7/2} = 4$$

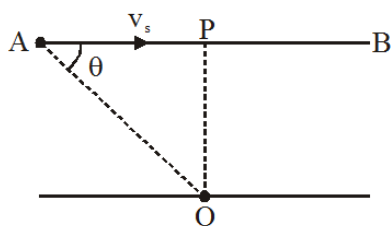
So accelerations of A & B is

$$|a_A| = |a_B| = \omega^2 \frac{L}{2} = 4^2 \times \frac{7}{2} = 56$$



6. Ans (2250.00)

The graph shows the situation shown in figure below. The observed frequency will initially be more than the natural frequency. When the source is at P, observed frequency is equal to its natural frequency i.e., 2000 Hz.



$$\text{For region AP : } f = f_0 \left(\frac{v}{v - v_s \cos \theta} \right)$$

$$\text{For PB : } f = f_0 \left(\frac{v}{v + v_s \cos \theta} \right)$$

Minimum value of f will be:

$$f_{\min} = f_0 \left(\frac{v}{v + v_s} \right) \text{ when } \cos \theta = 1$$

$$\text{or } 1800 = 2000 \left(\frac{300}{300 + v_s} \right)$$

Solving this we get, $v_s = 33.33 \text{ m/s}$

and maximum value of f can be

$$f_{\max} = f_0 \left(\frac{v}{v - v_s} \right) \text{ when } \cos \theta = 1$$

$$\text{or } f_{\max} = 2000 \left(\frac{300}{300 - 33.33} \right) = 2250 \text{ Hz}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (20)

All the elements of circuit are in parallel arrangement

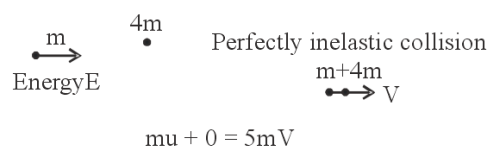
$$\frac{1}{R_{\text{eq}}} = \frac{1}{40} + \frac{1}{40} + \frac{1}{40} + \frac{1}{40} + \frac{1}{20} + \frac{1}{20} = \frac{4}{40} + \frac{2}{40}$$

$$R_{\text{eq}} = 5 \Omega$$

$$\text{Power} = V^2/R = 20 \text{ W}$$

8. Ans (6045)

For all collision to take place electron has to excite to $n = 3$.

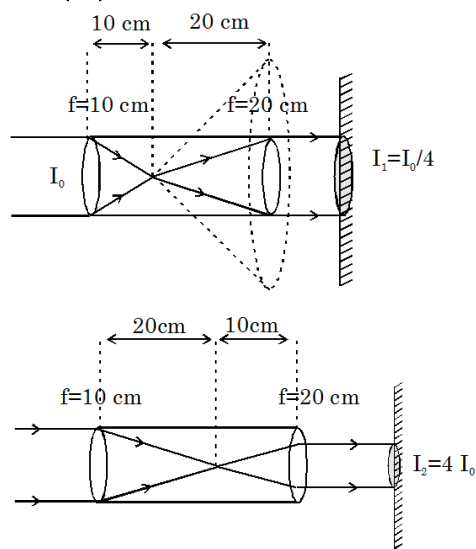


$$\text{Loss} = \frac{1}{2} \mu v^2 - \frac{1}{2} 5m \times \left(\frac{4}{5} \right)^2 = \frac{4E}{5}$$

$$\frac{4E}{5} = 12.09 \times (2)^2 \text{ eV}$$

$$\Rightarrow E = 60.45 \text{ eV}$$

9. Ans (4)



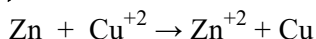
$$\therefore \frac{I_2}{I_1} = 16 ;$$

$$\therefore N = 4$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B,C,D)**



Initial - 0.9 M 0.1 M -

Discharge - 0.5 M 0.5 M -

Re-charge 0.8 M 0.2 M

(Reaction is reversed during charging)

$$E = 1.1 - \frac{0.06}{2} \log \frac{1}{4} = 1.1 + \frac{0.06}{2} \times 2 \times 0.30 = 1.118 \text{ V}$$

2. **Ans (A,C)**

Ag_2S and CuFeS_2 are sulphide ores and froth flotation is used for the concentration of the ore.

3. **Ans (A,B,C,D)**

For cyclic process $\Delta H = \Delta U = 0$

$$\Delta U_{\text{total}} = q_{\text{total}} + w_{\text{total}}$$

$$q_{\text{total}} = -w_{\text{total}}$$

$$w_{\text{total}} = w_{AB} + w_{BC} + w_{CA}$$

$$= \left[-2 + 0 + 0.5 \times 2 + \frac{1}{2} \times 0.5 \times 2 \right]$$

$$= -0.5$$

$$= -50.6625 \text{ J} \Rightarrow q_{\text{total}} = 50.6625 \text{ J}$$

4. **Ans (B,C,D)**

$$\Delta H_{\text{fusion}} \text{ for } 900 \text{ g} = \frac{5.99}{18} \times 900 = 299.5 \text{ kJ}$$

Hence choice (D) is correct.

ΔS_{fusion} for 900 g of ice

$$= \frac{\Delta H_{\text{fusion}}}{T_{\text{fusion}}} = \frac{299.5}{273} = 1.097 \text{ kJ K}^{-1} = 1097 \text{ JK}^{-1}$$

Hence choice (B) is correct.

ΔS_{fusion} for liquid water when it freezes

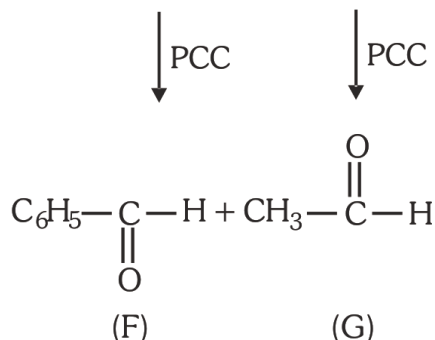
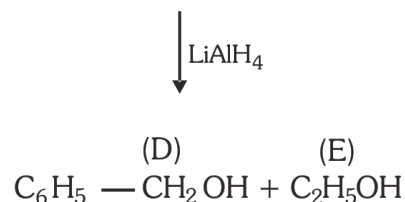
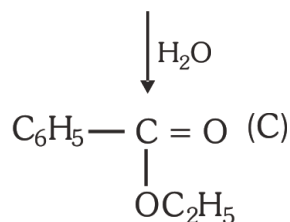
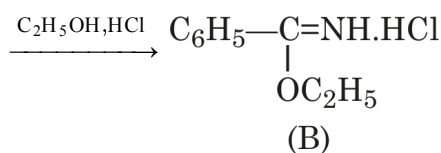
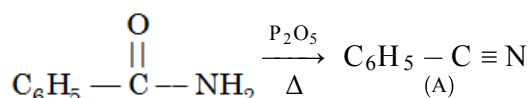
Here heat is liberated

$$\Delta H_{\text{fus}} = -5.99$$

$$\Delta S = \frac{-5.99}{273} \times 1000 = -21.94 \text{ JK}^{-1} \text{ mol}^{-1}$$

Hence choice (C) is correct. It follows that choice (A) is incorrect.

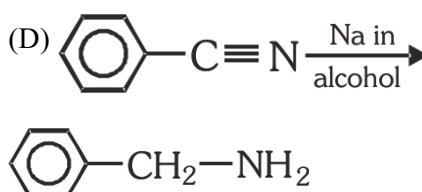
5. **Ans (B,C,D)**



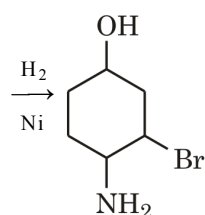
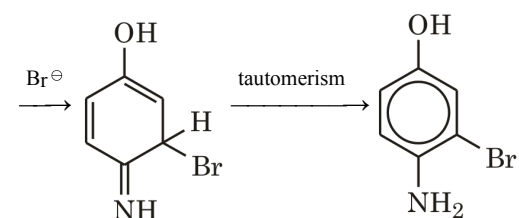
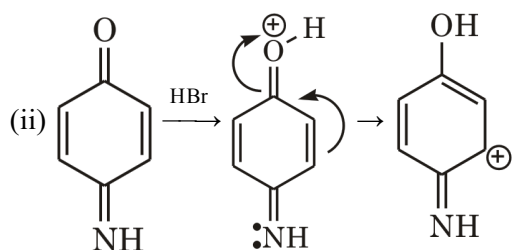
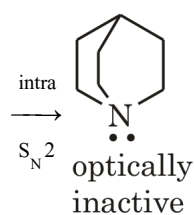
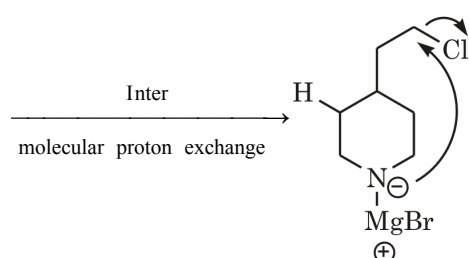
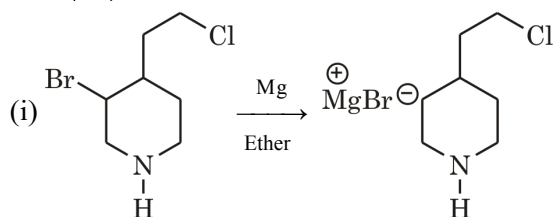
(A) (F) can give Cannizzaro reaction, (G) cannot undergoes Cannizzaro in the presence of conc. NaOH.

(B) (G) can give iodoform test, F cannot.

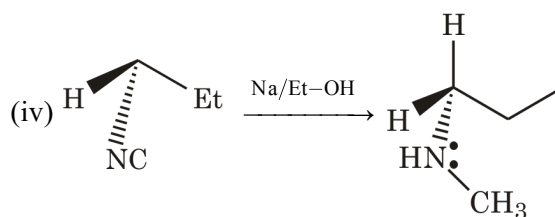
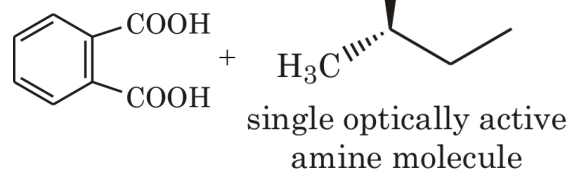
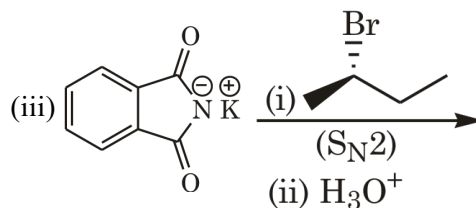
(C) all aldehydes give rise to a crystalline solid with 2, 4, -DNP.



6. Ans (C)



Multiple stereoisomers, optically active of amine as 3 chiral centres are generated.



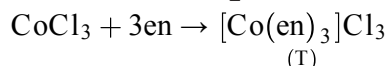
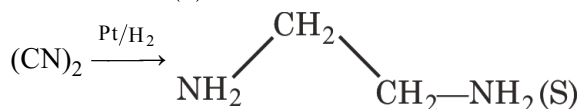
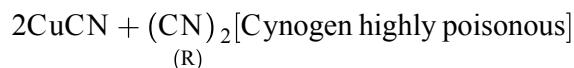
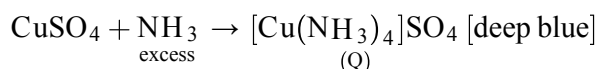
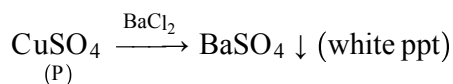
This is optically inactive 2° amine due to umbrella inversion at N.

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (B)

Solution (7 to 8)



S is bidentate ligand & R is cynogen.

∴ Option (B) is correct.

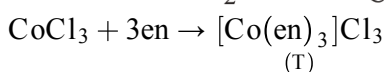
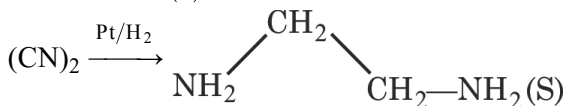
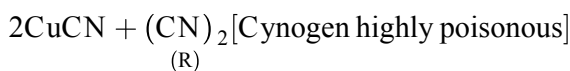
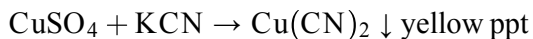
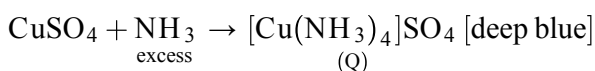
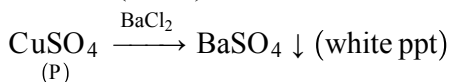
Q is $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ & T is $[\text{Co}(\text{en})_3]\text{Cl}_3$

Hence A, B and C all the options are correct.

∴ (D) is the correct options.

8. **Ans (D)**

Solution (7 to 8)



S is bidentate ligand & R is cynogen.

∴ Option (B) is correct.

Q is $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ & T is $[\text{Co}(\text{en})_3]\text{Cl}_3$

Hence A, B and C all the options are correct.

∴ (D) is the correct options.

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (564.70 to 565.00)**

(11 to 12)

The given data are

$$P_A^0 = 300 \text{ torr} \quad P_B^0 = 800 \text{ torr} \quad y_A = 0.25$$

$$\text{Hence } Y_B = 0.75$$

On calculation using enrichment formula, we get mol fraction of A in first condensate

$$\text{drop} = X_A = 0.47$$

$$\text{Now, Using R.L. } P = X_A P_A^0 + (1 - X_A) P_B^0$$

$$\text{We get } 0.47 \times 300 + 0.53 \times 800 = 565 \text{ torr}$$

When almost entire vapour mixture is compressed, then for last bubble of vapour

$$X_{A(\text{final})} = 0.25 \text{ and } X_{B(\text{final})} = 0.75 \text{ (same as mol fractions in ORIGINAL vapor mixture)}$$

$$\text{Then } X_A P_A^0 + X_B P_B^0 = 0.25 \times 300 + 0.75 \times 800 = 675 \text{ torr}$$

2. **Ans (675.00)**

(11 to 12)

The given data are

$$P_A^0 = 300 \text{ torr} \quad P_B^0 = 800 \text{ torr} \quad y_A = 0.25$$

$$\text{Hence } Y_B = 0.75$$

On calculation using enrichment formula, we get mol fraction of A in first condensate
drop = $X_A = 0.47$

$$\text{Now, Using R.L. } P = X_A P_A^0 + (1 - X_A) P_B^0$$

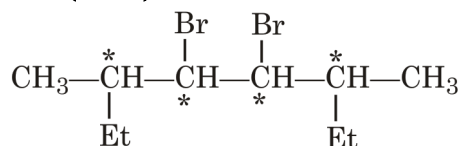
$$\text{We get } 0.47 \times 300 + 0.53 \times 800 = 565 \text{ torr}$$

When almost entire vapour mixture is compressed, then for last bubble of vapour

$$X_{A(\text{final})} = 0.25 \text{ and } X_{B(\text{final})} = 0.75 \text{ (same as mol fractions in ORIGINAL vapor mixture)}$$

$$\text{Then } X_A P_A^0 + X_B P_B^0 = 0.25 \times 300 + 0.75 \times 800 = 675 \text{ torr}$$

4. **Ans (4.00)**



5. **Ans (13.00)**

(15 to 16)

$$X \text{ is (Trp - Trp)} \Rightarrow \text{DBE} = 4 + 2 + 1 + 2 + 4 = 13$$

$$Y \text{ is (Met - Met)} \Rightarrow \text{DBE} = 1$$

6. **Ans (1.00)**

(15 to 16)

$$X \text{ is (Trp - Trp)} \Rightarrow \text{DBE} = 4 + 2 + 1 + 2 + 4 = 13$$

$$Y \text{ is (Met - Met)} \Rightarrow \text{DBE} = 1$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (335)**

x = 3 (a, d, f give +ve iodoform test)

y = 3 (d, c, e give +ve tollen's test)

z = 5 (a, b, g, h, i give +ve NaHCO_3 test)

8. **Ans (6)**

A buffer solution should contain weak acid and its conjugate base or weak base and its conjugate acid. Except (i), all combination can form buffer.

9. **Ans (8)**

Intermolecular hydrogen bonding causes increase in boiling point.

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

$f'(x)$ must be of least degree so take

$$f'(x) = a(x-1)(x-3)$$

$$\Rightarrow f(x) = \frac{a}{3}x^3 - 2ax^2 + 3ax + c$$

Using $f(1) = 6$ and $f(3) = 2$ we get $a = 3$, $c = 2$

$$\therefore f(x) = x^3 - 6x^2 + 9x + 2$$

2. Ans (A,D)

$$\frac{x^2}{16} + \frac{y^2}{1} = 1$$

$$a^2 = 16, b^2 = 1$$

$$a = 4, b = 1$$

let $A(x_1, y_1)$ and $B(x_2, y_2)$ then P, Q will be

$$P\left(x_1, \frac{y_1}{4}\right) \text{ and } Q\left(x_2, \frac{y_2}{4}\right)$$

$$\text{mid point of AB will be } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

mid point of PQ will be

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{8}\right) = (h, k)$$

$$\Rightarrow \text{mid point of AB will be } (h, 4k)$$

let point $S(t, t+8)$ be a point on $y = x + 8$

AB is chord of contact of $x^2 + y^2 = 16$

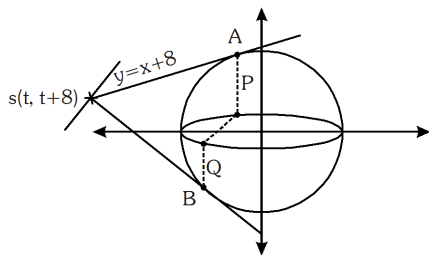
$$\Rightarrow xt + y(t+8) = 16 \quad \dots(i)$$

AB as chord whose mid point is $(h, 4k)$

$$xh + y(4k) = h^2 + 16k^2 \quad \dots(ii)$$

$$\frac{t}{h} = \frac{t+8}{4k} = \frac{16}{h^2 + 16k^2}$$

$$\Rightarrow t = \frac{16h}{h^2 + 16k^2} \text{ and } t+8 = \frac{64k}{h^2 + 16k^2}$$



$$\Rightarrow \frac{16h}{h^2 + 16k^2} + 8 = \frac{64k}{h^2 + 16k^2}$$

$$\Rightarrow 2h + (h^2 + 16k^2) = 8k$$

$$\Rightarrow x^2 + 16y^2 + 2x - 8y = 0$$

$$x^2 + \alpha y^2 + \beta x + \gamma y = 0$$

$$\Rightarrow \alpha = 16, \beta = 2, \gamma = -8$$

$$\Rightarrow \alpha - \beta + \gamma = 6$$

3. Ans (A,B,C)

$$(A) \text{ Take } h(x) = e^{\beta x} \cdot f(x), x \in [a, b]$$

Use Rolle's theorem

$$0 = h'(a) = [\beta f(a) + f'(a)] e^{\beta a}$$

$$\Rightarrow \beta f(a) + f'(a) = 0$$

$$(B) \text{ Take } h(x) = e^{g(x)} \cdot f(x) \quad \forall x \in [a, b],$$

use Rolle's theorem

$$(C) \text{ Take, } h(x) = \frac{f(x)}{g(x)} \quad \forall x \in [a, b]$$

and apply Rolle's theorem

$$(D) \text{ Take } h(x) = (f(x))^2 - x^2 \quad \forall x \in [a, b]$$

and apply Rolle's theorem

4. Ans (B,D)

$$S = \sum_{r=0}^n (-1)^r {}^nC_r \frac{1+rx}{(1+nx)^r}$$

(let $n = 2018$, $x = \ln 2$)

$$= \sum_{r=0}^n (-1)^r \frac{{}^nC_r}{(1+nx)^r} + \sum_{r=0}^n (-1)^r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1} \frac{rx}{(1+nx)^r}$$

$$= \sum_{r=0}^n (-1)^r \frac{{}^nC_r}{(1+nx)^r} - \frac{nx}{(1+nx)} \sum_{r=0}^n (-1)^{r-1} {}^{n-1}C_{r-1} \frac{1}{(1+nx)^{r-1}}$$

$$= \left(1 - \frac{1}{1+nx}\right)^n - \frac{nx}{1+nx} \left(1 - \frac{1}{1+nx}\right)^{n-1}$$

$$= \left(\frac{nx}{1+nx}\right)^n - \frac{nx}{1+nx} \cdot \left(\frac{nx}{1+nx}\right)^{n-1} = 0$$

5. Ans (B,C)

$$I_n = \int_0^{\frac{3\pi}{2}} (\ln |\sin x|) \cos(2nx) dx$$

applying integration by parts

$$I_n = \left\{ \ln |\sin x| \cdot \frac{\sin 2nx}{2n} \right\}_0^{\frac{3\pi}{2}} - \int_0^{\frac{3\pi}{2}} \frac{\cot x \cdot \sin 2nx}{2n} dx$$

$$I_n = 0 - \frac{1}{2n} I_n'$$

$$I_n' = \int_0^{\frac{3\pi}{2}} \frac{\cos x \cdot \sin 2nx}{\sin x} dx$$

$$I_n' - I_{n-1}' = \int_0^{\frac{3\pi}{2}} \frac{\cos x (\sin 2nx - \sin(2n-2)x)}{\sin x} dx$$

$$= \int_0^{\frac{3\pi}{2}} \frac{2 \cos x \cdot \cos(2n-1)x \sin x}{\sin x} dx$$

$$I_n' - I_{n-1}' = \int_0^{\frac{3\pi}{2}} 2 \cos(2n-1)x \cdot \cos x dx = 0$$

$$I_n' = I_{n-1}' = I_{n-2}' = \dots = I_1'$$

$$I_1' = \int_0^{\frac{3\pi}{2}} \frac{\sin 2x \cos x}{\sin x} dx$$

$$= \int_0^{\frac{3\pi}{2}} 2 \cos^2 x dx = \int_0^{\frac{3\pi}{2}} (1 + \cos 2x) dx$$

$$= \frac{3\pi}{2}$$

$$I_n = -\frac{3\pi}{4n}$$

$$12I_3 = -3\pi$$

$$16I_2 = -6\pi$$

$$\Rightarrow 12I_3 - 16I_2 = 3\pi$$

6. Ans (A,C)

$$f(x) = g\left(\frac{1}{x^5}\right) e^{-\frac{1}{x^6}}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{g\left(\frac{1}{x^5}\right)}{e^{\frac{1}{x^6}}}$$

$$= \lim_{t \rightarrow \infty} \frac{g(t^5)}{e^{t^6}} = 0$$

$$f'(x) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{g\left(\frac{1}{x^5}\right) e^{-\frac{1}{x^6}}}{x} = \lim_{x \rightarrow 0} \frac{g\left(\frac{1}{x^5}\right)}{x e^{\frac{1}{x^6}}}$$

$$= \lim_{t \rightarrow \infty} \frac{t g(t^5)}{e^{t^6}} = 0$$

$\Rightarrow f(x)$ is continuous and differentiable at $x = 0$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

Tangent:

$$Y - y = \frac{dy}{dx}(X - x). \text{ So } OP = x - y \frac{dx}{dy}, OQ = y - x \frac{dy}{dx}$$

Now, if $A = 1$ i.e.,

$$x - y \frac{dx}{dy} + y - x \frac{dy}{dx} = 2 \Rightarrow x - \frac{y}{y'} + y - xy' = 2$$

...(1)

Differentiating we get,

$$1 + \frac{y}{(y')^2} y'' - 1 + y' - y' - xy'' = 0$$

$$\Rightarrow y'' \left(\frac{y}{(y')^2} - x \right) = 0.$$

Taking $y'' = 0 \Rightarrow y' = k$ (a constant)

Putting in equation (1), we get,

$$x - \frac{y}{k} + y - xk = 2 \Rightarrow x = \frac{y}{k} + \frac{2}{1-k}$$

as the desired solution.

8. **Ans (C)**

$$\text{If } G = 1 \text{ then } \left(x - y \frac{dx}{dy}\right) \left(y - x \frac{dy}{dx}\right) = 1$$

$$\Rightarrow 2xy - x^2 \frac{dy}{dx} - y^2 \frac{dx}{dy} = 1 \Rightarrow x^2 y' - 2xy + \frac{y^2}{y'} + 1 = 0$$

... (1)

Differentiating we get,

$$x^2 y'' + 2xy' - 2xy' - 2y + 2y - \frac{y^2}{(y')^2} \cdot y'' = 0$$

$$\Rightarrow y'' \left(x^2 - \frac{y^2}{(y')^2}\right) = 0. \text{ Taking } y'' = 0 \Rightarrow y' = k,$$

(a constant).

Putting in equation (1), we get,

$$kx^2 - 2xy + \frac{1}{k}y^2 + 1 = 0 \text{ as the desired solution.}$$

9. **Ans (C)**

$$\frac{y''}{y'} = \frac{2y'}{y} \Rightarrow \frac{y'}{y^2} = k$$

$$\int \frac{dy}{y^2} = \int k dx \Rightarrow -\frac{1}{y} = kx + \lambda \quad \dots(1)$$

since (2,2) and $\left(8, \frac{1}{2}\right)$ lies on (1), we get $C_1 :$

$$xy = 4$$

shortest distance between

$$C_1 : xy = 4 \text{ and } C_2 : x^2 + y^2 = 4 \text{ is}$$

$$(2\sqrt{2} - 2) = (\sqrt{8} - 2)$$

$$\text{Hence, } (a + b) = 10$$

10. **Ans (D)**

$$\frac{y''}{y'} = \frac{2y'}{y} \Rightarrow \frac{y'}{y^2} = k$$

$$\int \frac{dy}{y^2} = \int k dx \Rightarrow -\frac{1}{y} = kx + \lambda \quad \dots(1)$$

since (2,2) and $\left(8, \frac{1}{2}\right)$ lies on (1), we get $C_1 :$

$$xy = 4$$

shortest distance between

$$C_1 : xy = 4 \text{ and } C_2 : x^2 + y^2 = 4 \text{ is}$$

$$(2\sqrt{2} - 2) = (\sqrt{8} - 2)$$

$$\text{Hence, } (a + b) = 10$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (47.00)**

Required probability

$$= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^7 = \frac{47}{2^7}$$

2. **Ans (34.00)**

There are five cases 0, 1, 2, 3, or 4 wins

$$\text{Required probability} = \frac{1+7+15+10+1}{2^7} = \frac{34}{2^7}$$

3. **Ans (1.50)**

$$g(x) = x - k \text{ where } k = \int_0^1 f(t) dt \quad \dots(1)$$

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x (t - k) dt$$

$$f(x) = \frac{x^3}{2} + 1 - x \left(\frac{x^2}{2} - kx\right)$$

$$f(x) = kx^2 + 1 \quad \dots(2)$$

$$\text{Now from (1) } k = \int_0^1 (kt^2 + 1) dt$$

$$k = \frac{k}{3} + 1 \Rightarrow k = \frac{3}{2}$$

$$\Rightarrow \int_0^1 f(t) dt = k = \frac{3}{2} = 1.50$$

4. **Ans (1.00)**

$$g(x) = x - k \text{ where } k = \int_0^1 f(t) dt \quad \dots(1)$$

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x (t - k) dt$$

$$f(x) = \frac{x^3}{2} + 1 - x \left(\frac{x^2}{2} - kx\right)$$

$$f(x) = kx^2 + 1 \quad \dots(2)$$

$$\text{Now from (1) } k = \int_0^1 (kt^2 + 1) dt$$

$$k = \frac{k}{3} + 1 \Rightarrow k = \frac{3}{2}$$

$$f(x) = \frac{3}{2}x^2 + 1 ; g(x) = x - \frac{3}{2}$$

$$f'(x_1) = 1 \Rightarrow 3x_1 = 1$$

5. Ans (0.00)

$$a_{ij} = (k)^{i+j}$$

$$A = \begin{bmatrix} k^2 & k^3 & k^4 \\ k^3 & k^4 & k^5 \\ k^4 & k^5 & k^6 \end{bmatrix}$$

$$|A| = k^{12} (1 + 1 + 1 - 1 - 1 - 1) = 0$$

6. Ans (3.00)

$$A^2 = 3A$$

$$A^2 = \begin{bmatrix} k^2 & k^3 & k^4 \\ k^3 & k^4 & k^5 \\ k^4 & k^5 & k^6 \end{bmatrix} \begin{bmatrix} k^2 & k^3 & k^4 \\ k^3 & k^4 & k^5 \\ k^4 & k^5 & k^6 \end{bmatrix} = \begin{bmatrix} k^4 + k^6 + k^8 & k^5 + k^7 + k^9 & k^6 + k^8 + k^{10} \\ k^5 + k^7 + k^9 & k^6 + k^8 + k^{10} & k^7 + k^9 + k^{11} \\ k^6 + k^8 + k^{10} & k^7 + k^9 + k^{11} & k^8 + k^{10} + k^{12} \end{bmatrix}$$

$$\Rightarrow A^2 = k^2 (1 + k^2 + k^4) \begin{bmatrix} k^2 & k^3 & k^4 \\ k^3 & k^4 & k^5 \\ k^4 & k^5 & k^6 \end{bmatrix}$$

$$3A = 3 \begin{bmatrix} k^2 & k^3 & k^4 \\ k^3 & k^4 & k^5 \\ k^4 & k^5 & k^6 \end{bmatrix}$$

$$\text{Hence } k^2(k^4 + k^2 + 1) = 3 \text{ or } k = 0 \Rightarrow k = 0, \pm 1$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (7)

Let equation of new plane $2x - 2y + z - 3 + \lambda z = 0$

Point (3, 1, 1) lie on it $\Rightarrow \lambda = -2$

Hence equation of new plane $2x - 2y - z = 3$

$$\cos \alpha = \frac{4 + 4 - 1}{3.3} = \frac{7}{9}$$

8. Ans (683)

$$\text{Given } a_{n+1} = a_n + 4n + 3$$

Now

$$a_k - a_1 = \sum_{n=1}^{k-1} (a_{n+1} - a_n) = \sum_{n=1}^{k-1} (4n + 3) = 4(k-1) \frac{(k)}{2} + 3(k-1) = 2k^2 + k - 3$$

$$\Rightarrow a_k = 2k^2 + k - 3$$

So,

$$\lim_{k \rightarrow \infty} \frac{\sqrt{2k^2 + k - 3} + \sqrt{2(4k)^2 + (4k) - 3} + \dots + \sqrt{2(4^{10}k)^2 + (4^{10}k) - 3}}{\sqrt{2k^2 + k - 3} + \sqrt{2(2k)^2 + (2k) - 3} + \dots + \sqrt{2(2^{10}k)^2 + (2^{10}k) - 3}}$$

$$= \lim_{k \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{k} - \frac{3}{k^2}} + \sqrt{2(4)^2 + \frac{4}{k} - \frac{3}{k^2}} + \dots + \sqrt{2(4^{10})^2 + \frac{4^{10}}{k} - \frac{3}{k^2}}}{\sqrt{2 + \frac{1}{k} - \frac{3}{k^2}} + \sqrt{2(2)^2 + \frac{2}{k} - \frac{3}{k^2}} + \dots + \sqrt{2(2^{10})^2 + \frac{2^{10}}{k} - \frac{3}{k^2}}}$$

$$= \frac{\sqrt{2} + \sqrt{2(4^2)} + \dots + \sqrt{2(4^{10})^2}}{\sqrt{2} + \sqrt{2(2^2)} + \dots + \sqrt{2(2^{10})^2}} = \frac{\sqrt{2}(1 + 4 + 4^2 + \dots + 4^{10})}{\sqrt{2}(1 + 2 + 2^2 + \dots + 2^{10})} = \frac{\frac{1}{3}(4^{11} - 1)}{2^{11} - 1} = 683$$

9. Ans (5)

$$f(x) = \sqrt{\cos^{-1}(2x) + 2} + \sqrt{1 - \cos^{-1}(2x)}$$

$$1 - \cos^{-1}(2x) \geq 0$$

$$\Rightarrow -1 \leq 2x \leq 1 \quad \cos^{-1}(2x) \leq 1$$

$$\Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2} \quad 2x \geq \cos(1)$$

$$x \geq \frac{1}{2} \cos 1$$

$$\text{Domain is } \frac{1}{2} \cos 1 \leq x \leq \frac{1}{2}$$

$$\& \text{ Range : } [\sqrt{3}, \sqrt{2} + 1]$$

$$a + b = 5$$